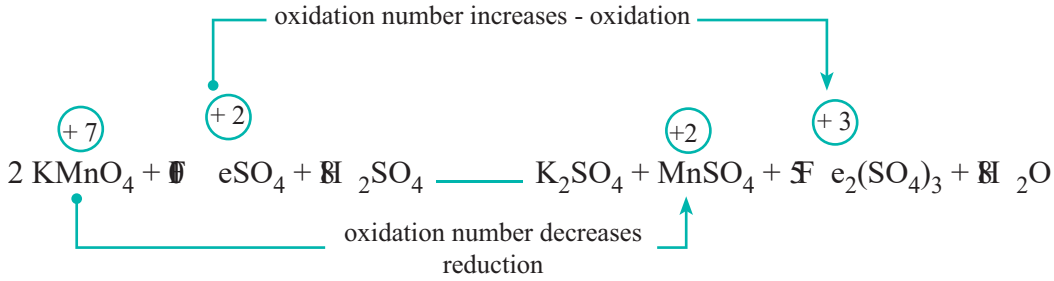
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Redox reactions in terms of oxidation numbers

During redox reactions, the oxidation number of elements changes. A reaction in which oxidation number of the element increases is called oxidation. A reaction in which it decreases is called reduction.

Consider the following reaction



In this reaction, manganese in potassium permanganate ($KMnO_4$) favours the oxidation of ferrous sulphate ($FeSO_4$) into ferric sulphate ($Fe_2(SO_4)_3$ by gaining electrons and thereby gets reduced. Such reagents are called oxidising agents or oxidants. Similarly , the reagents which facilitate reduction by releasing electrons and get oxidised are called reducing agents.

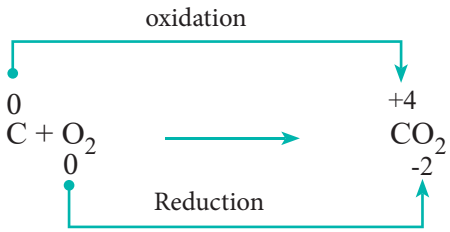
1.8.2 Types of Redox Reactions

Redox reactions are classified into the following types.

1. Combination reactions:

Redox reactions in which two substances combine to form a single compound are called combination reaction.

Example:



2. Decomposition reaction:

Redox reactions in which a compound breaks down into two or more components are called decomposition reactions. These reactions are opposite to combination reactions. In these reactions, the oxidation number of the different elements in the same substance is changed.